

KATHAQUEST TEST CHAPTER

Our Solar System: Worlds Around the Sun

Meet the Sun, rocky planets, giant planets and smaller space objects



Not to scale: planets orbit the Sun at different distances

Recommended reading level: Ages 8-12

Original educational content created for product testing

What you will learn

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| 1 | Describe the Sun's role in the solar system. |
| 2 | Group planets as rocky worlds or giant planets. |
| 3 | Explain the difference between rotation and revolution. |
| 4 | Identify smaller objects such as moons, asteroids and comets. |

Key words: star, planet, orbit, rotation, revolution, moon, asteroid, comet



Not to scale: planets orbit the Sun at different distances

1. The Sun at the centre

The solar system is our neighbourhood in space. At its centre is the **Sun**, a star made mostly of extremely hot hydrogen and helium. It gives Earth light and heat.

The Sun contains almost all the mass in the solar system. Its gravity keeps planets, dwarf planets, asteroids and comets moving in curved paths called **orbits**.

The Sun is not the largest star, but it looks large because it is much closer to Earth than other stars.

2. The four rocky planets

Mercury, Venus, Earth and Mars are the inner planets. They are called rocky planets because they have solid surfaces. Mercury is closest to the Sun. Venus has a thick atmosphere that traps heat. Earth has liquid surface water and known life. Mars is cold and dusty, with giant volcanoes and canyons.

Although pictures often place the planets close together, the real distances are enormous. Most of the solar system is empty space.

Visual moment to look for: a size comparison of Mercury, Venus, Earth and Mars.

3. The giant planets

Jupiter and Saturn are gas giants made mainly of hydrogen and helium. Jupiter is the largest planet and has a long-lasting storm called the Great Red Spot. Saturn is famous for rings made of ice, dust and

rock.

Uranus and Neptune are often called ice giants because they contain more water, ammonia and methane compounds. Both are very cold and have powerful winds.

Visual moment to look for: Saturn's rings or Jupiter's swirling clouds.

4. Days, years and seasons

A planet spins around an imaginary line called its axis. This movement is **rotation**. Earth takes about 24 hours to rotate once, producing day and night.

A planet also travels around the Sun. This movement is **revolution**. Earth takes about 365 days to complete one orbit. Earth's tilted axis, combined with its revolution, causes the seasons.

Do not confuse rotation with revolution: spin in place versus travel around another object.

5. Moons and smaller objects

Many planets have natural satellites called moons. Earth has one Moon, while some giant planets have dozens. Moons can have mountains, craters, ice and even hidden oceans.

Asteroids are rocky objects, many of which orbit between Mars and Jupiter. Comets contain ice and dust. When a comet approaches the Sun, heat can create a glowing cloud and a long tail.

A comet's tail points generally away from the Sun because of sunlight and the solar wind.

Chapter summary

- The Sun is the central star whose gravity holds the solar system together.
- The inner planets are rocky; the outer planets are giants.
- Rotation produces a planet's day, while revolution determines its year.
- Moons, asteroids and comets are important members of the solar system.

Glossary

Word	Meaning
star	A huge glowing ball of hot gas that produces energy.
planet	A large world that orbits a star and is nearly round.
orbit	The curved path of one object around another.
rotation	The spinning of an object around its axis.
revolution	The movement of one object around another.
comet	An icy object that may form a glowing cloud and tail near the Sun.

Quick check

1. Why do planets remain in orbit around the Sun?
2. Which four planets have rocky surfaces?
3. How are rotation and revolution different?
4. Why does Saturn look unusual?
5. What happens when a comet travels close to the Sun?

Testing note: This chapter is intentionally written with clear concepts, visual events and child-friendly language. It is suitable for testing chapter extraction, concept detection, media search, narration, quiz generation and revision-video workflows.